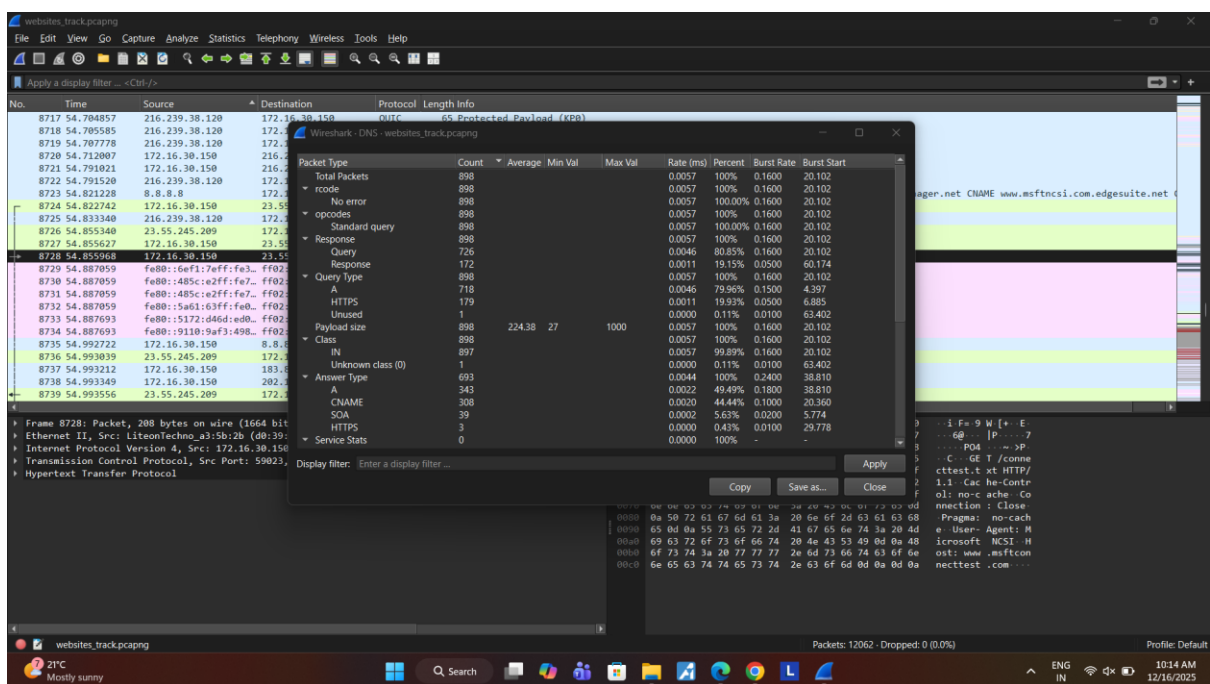


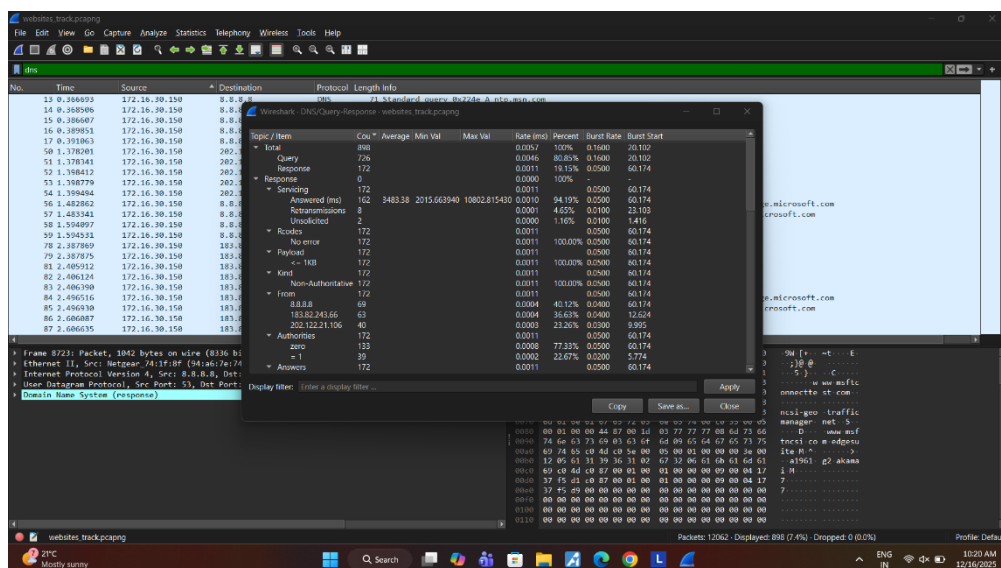
Wireshark to investigate application protocols in operation

Task 1:

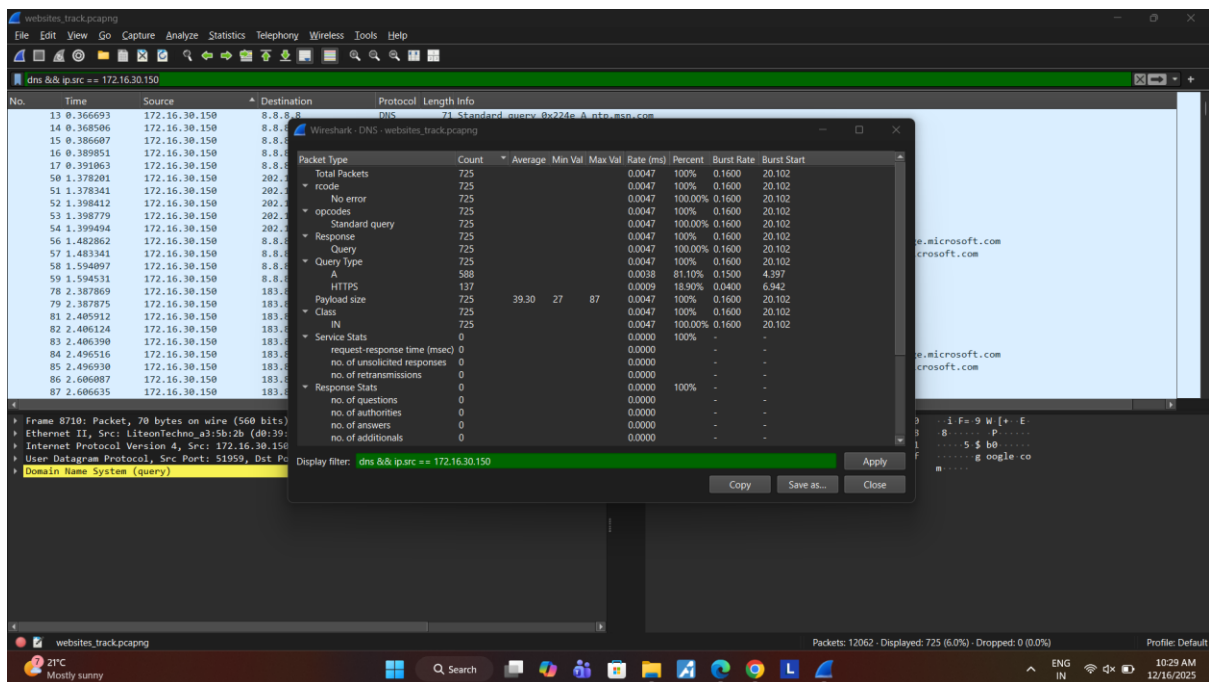
How many DNS queries are sent from your browser host machine to DNS Servers



How many DNS servers are involved :



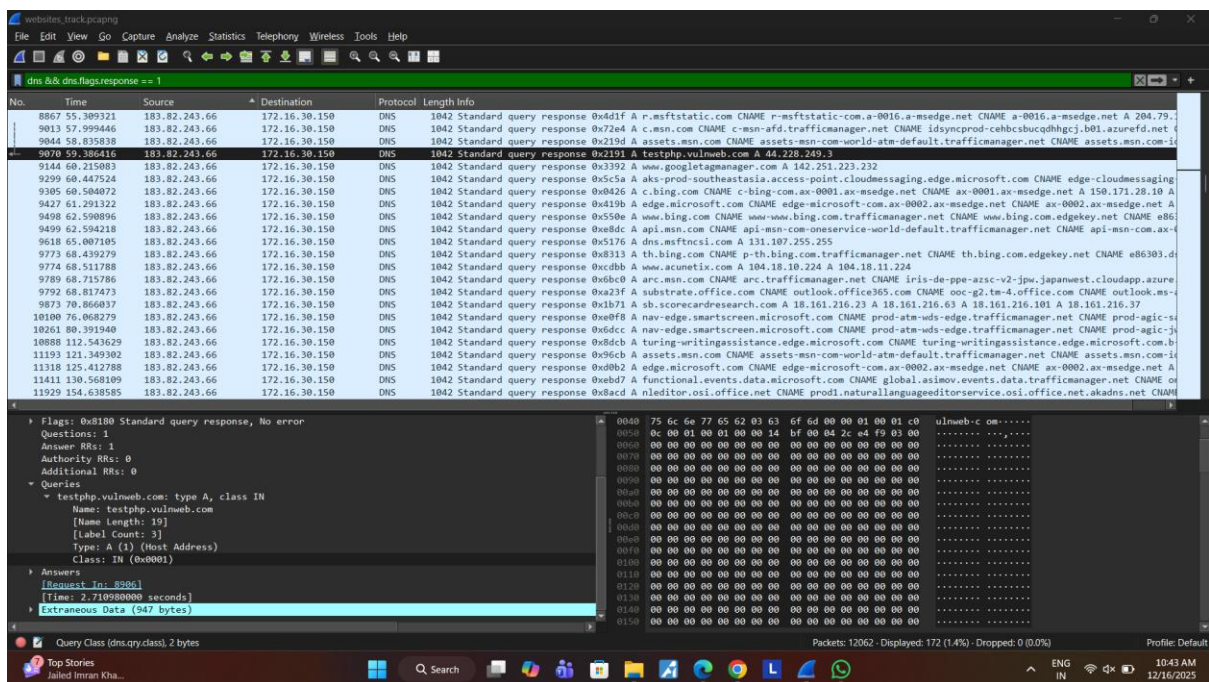
Which DNS Server replies with actual IP Addresses:



Do all DNS servers respond

Ans: NO

Clearly list the resource records involved in resolving the IP address of the site, mentioning, Name, value, type, TTL appropriately in the complete resolving process of this DNS conversation including query/queries and response/answer(s).



2. How many HP requests (Type and respective count of requests), responses (status code and phrase of each of the responses) did the browser send and receive :

Wireshark packet capture analysis of HTTP traffic. The packet list shows a GET request for /style.css. The packet details pane shows the request structure. The packet bytes pane shows the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
9165	60.281943	172.16.30.150	44.228.249.3	HTTP	411	GET /style.css HTTP/1.1
9166	60.282458	172.16.30.150	44.228.249.3	Wireshark: HTTP / Packet Counter - websites_track.pcapng		
9439	61.539985	172.16.30.150	44.228.249.3			
9536	63.800126	172.16.30.150	44.228.249.3			
10907	112.668700	172.16.30.150	44.228.249.3			
10947	113.040783	172.16.30.150	44.228.249.3			
11130	119.836620	172.16.30.150	44.228.249.3			
11463	133.658948	172.16.30.150	44.228.249.3			
11480	133.961928	172.16.30.150	44.228.249.3			
11816	150.610300	172.16.30.150	44.228.249.3			
11827	150.956638	172.16.30.150	44.228.249.3			
8739	14.933556	23.55.245.209	172.16.30.150			
9124	60.173616	44.228.249.3	172.16.30.150			
9327	60.523650	44.228.249.3	172.16.30.150			
9331	60.529616	44.228.249.3	172.16.30.150			
9477	62.029880	44.228.249.3	172.16.30.150			
9542	64.037841	44.228.249.3	172.16.30.150			
10943	112.954438	44.228.249.3	172.16.30.150			
10959	113.364871	44.228.249.3	172.16.30.150			
11153	120.122469	44.228.249.3	172.16.30.150			
11477	133.945606	44.228.249.3	172.16.30.150			
11484	134.253478	44.228.249.3	172.16.30.150			
11826	150.944267	44.228.249.3	172.16.30.150			

Packet Type: Total HTTP Packets: 72, HTTP Request Packets: 72, SEARCH: 59, GET: 10, POST: 3, Other HTTP Packets: 0, HTTP Response Packets: 0, 200 OK: 0, 3xx Redirection: 0, 4xx Client Error: 0, 5xx Server Error: 0, 1xx Informational: 0.

Display filter: http.request

Wireshark packet capture analysis of HTTP traffic. The packet list shows a GET request for /style.css. The packet details pane shows the request structure. The packet bytes pane shows the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
9165	60.281943	172.16.30.150	44.228.249.3	HTTP	411	GET /style.css HTTP/1.1
9166	60.282458	172.16.30.150	44.228.249.3	Wireshark: HTTP / Packet Counter - websites_track.pcapng		
9439	61.539985	172.16.30.150	44.228.249.3			
9536	63.800126	172.16.30.150	44.228.249.3			
10907	112.668700	172.16.30.150	44.228.249.3			
10947	113.040783	172.16.30.150	44.228.249.3			
11130	119.836620	172.16.30.150	44.228.249.3			
11463	133.658948	172.16.30.150	44.228.249.3			
11480	133.961928	172.16.30.150	44.228.249.3			
11816	150.610300	172.16.30.150	44.228.249.3			
11827	150.956638	172.16.30.150	44.228.249.3			
8739	14.933556	23.55.245.209	172.16.30.150			
9124	60.173616	44.228.249.3	172.16.30.150			
9327	60.523650	44.228.249.3	172.16.30.150			
9331	60.529616	44.228.249.3	172.16.30.150			
9477	62.029880	44.228.249.3	172.16.30.150			
9542	64.037841	44.228.249.3	172.16.30.150			
10943	112.954438	44.228.249.3	172.16.30.150			
10959	113.364871	44.228.249.3	172.16.30.150			
11153	120.122469	44.228.249.3	172.16.30.150			
11477	133.945606	44.228.249.3	172.16.30.150			
11484	134.253478	44.228.249.3	172.16.30.150			
11826	150.944267	44.228.249.3	172.16.30.150			

Packet Type: Total HTTP Packets: 88, HTTP Request Packets: 75, SEARCH: 59, GET: 10, POST: 3, NOTIFY: 1, Other HTTP Packets: 0, HTTP Response Packets: 13, 200 OK: 11, 3xx Redirection: 2, 4xx Client Error: 0, 5xx Server Error: 0, 1xx Informational: 0.

Display filter: Enter a display filter...

3. How many TCP Connections has the browser established overall:

The image shows a Wireshark packet capture of a TCP connection. The top pane displays a list of packets, with packet 213 selected. The middle pane shows the details of the selected packet, which is a TCP segment. The bottom pane shows the raw packet data in hexadecimal and ASCII.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
213	2025-12-16 12:59:51.711807	172.16.31.158	142.251.223.3	TCP	66	57066 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM

Packet Details:

Frame 213: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface Device\NPF...
Ethernet II, Src: AzureWavetec_08:00:0e:46:3d:47, Dst: Sonicwall_ee:46:3d:18:b1:69:ee:46:3d
Internet Protocol Version 4, Src: 172.16.31.158, Dst: 142.251.223.3
Transmission Control Protocol, Src Port: 57066, Dst Port: 443, Seq: 0, Len: 0
Source Port: 57066
Destination Port: 443
[Stream index: 8]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 1153280915
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1000 --- Header Length: 32 bytes (8)
Flags: 0x002 (SYN)
Window: 65535
[Calculated window size: 65535]
Checksum: 0x646c [Unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), No
[Timestamps]

The image shows the Wireshark Conversations window, which displays a list of all captured conversations. The table shows the source and destination addresses, ports, and the number of packets and bytes transferred for each conversation.

Address A	Port A	Address B	Port B	Packets	Bytes	Stream ID	Packets B → A	Bytes B → A	Packets A → B	Bytes A → B	Packets A → B	Bytes A → B	Bit Start	Duration	Bits/s A → B	Bits/s B → A
172.16.31.150	57073	4.144.165.14	443	58	25 kb	24	28	10 kb	30	14 kb	24311233	30.8631	2702	2702	2702	3686
172.16.31.150	57073	4.213.25.242	443	3	434 bytes	55	2	209 bytes	1	225 bytes	52000363	0.0882	18	18	18	20
172.16.31.150	50095	4.213.133.109	443	81	21 kb	73	39	8 kb	42	13 kb	72614425	77.9541	847	847	847	124
172.16.31.150	50729	13.89.179.14	443	90	36 kb	5	51	26 kb	39	10 kb	15.900744	39.3420	5292	5292	5292	2074
172.16.31.150	60971	13.89.179.14	443	21	10 kb	11	10	3 kb	11	7 kb	16.243623	39.0818	632	632	632	1483
172.16.31.150	61276	13.89.179.14	443	15	10 kb	10	7	3 kb	8	7 kb	15.981029	1.1081	19	19	19	49
172.16.31.150	60048	13.89.179.14	443	16	10 kb	8	8	3 kb	8	7 kb	15.980321	0.8329	28	28	28	66
172.16.31.150	51920	13.1076.158	443	32	12 kb	84	15	4 kb	17	8 kb	112.546160	4.4103	7214	7214	7214	13 kb
172.16.31.150	49533	13.1072.246.58	443	33	13 kb	76	15	3 kb	18	9 kb	75.261900	75.3431	356	356	356	988
172.16.31.150	51767	18.161.216.23	443	24	11 kb	72	10	2 kb	14	8 kb	69.025831	64.6403	302	302	302	1042
172.16.31.150	57095	20.42.65.84	443	16	10 kb	68	7	3 kb	9	7 kb	60.848017	1.1121	18	18	18	42
172.16.31.150	58938	20.42.65.84	443	86	30 kb	64	46	20 kb	40	10 kb	60.187298	95.5229	1650	1650	1650	851
172.16.31.150	63725	20.190.145.140	443	3	168 bytes	80	2	108 bytes	1	60 bytes	96.736029	0.1399	6173	6173	3429	3429
172.16.31.150	63726	20.190.145.140	443	3	168 bytes	81	2	108 bytes	1	60 bytes	96.736029	0.1405	6149	6149	3416	3416
172.16.31.150	49043	23.11.215.11	443	24	8 kb	43	11	2 kb	13	6 kb	37.099737	16.1061	1217	1217	2834	2834
172.16.31.150	52164	23.11.215.11	443	21	8 kb	47	10	2 kb	11	5 kb	37.100416	17.9577	1048	1048	2371	2371
172.16.31.150	55731	23.11.215.11	443	22	8 kb	46	10	2 kb	12	5 kb	37.100302	16.1045	1216	1216	2867	2867
172.16.31.150	58519	23.11.215.11	443	42	25 kb	45	15	4 kb	27	21 kb	37.100166	17.9729	3589	3589	9537	9537
172.16.31.150	58705	23.11.215.11	443	18	8 kb	44	8	2 kb	10	5 kb	37.100012	0.0344	529	529	1215	1215
172.16.31.150	58878	23.11.215.11	443	24	8 kb	48	11	2 kb	13	6 kb	37.100550	16.1042	1217	1217	2843	2843
172.16.31.150	58220	23.11.215.11	443	706	724 kb	33	210	15 kb	488	709 kb	32.077418	22.1975	5566	5566	2554	2554
172.16.31.150	51086	23.11.215.106	443	23	8 kb	74	10	2 kb	13	5 kb	75.259863	37.2914	511	511	1164	1164
172.16.31.150	61527	23.11.215.106	443	24	8 kb	75	11	2 kb	13	5 kb	75.260488	37.2916	529	529	1164	1164
172.16.31.150	59023	23.55.245.209	80	12	1 kb	56	6	490 bytes	6	547 bytes	54.822742	0.2311	16	16	16	18
172.16.31.150	49244	23.60.109.213	443	25	11 kb	58	12	3 kb	13	8 kb	57.724508	90.1028	263	263	706	706
172.16.31.150	54042	23.62.12.35	443	22	7 kb	39	10	2 kb	12	5 kb	35.204956	10.2144	1855	1855	3621	3621
172.16.31.150	55839	23.62.12.35	443	21	7 kb	41	9	2 kb	12	5 kb	35.205333	10.2151	1838	1838	3621	3621
172.16.31.150	59637	23.62.12.35	443	23	7 kb	40	11	3 kb	12	5 kb	35.205282	19.8525	1015	1015	1857	1857
172.16.31.150	64703	23.62.12.35	443	22	7 kb	42	10	2 kb	12	5 kb	35.205747	19.8522	993	993	1870	1870
172.16.31.150	63962	40.104.66.226	443	3	168 bytes	83	2	108 bytes	1	60 bytes	97.318488	0.0696	12	12	12	689
172.16.31.150	63963	40.104.66.226	443	3	168 bytes	82	2	108 bytes	1	60 bytes	97.317403	0.0711	12	12	12	675
172.16.31.150	52942	44.228.249.3	80	26	12 kb	61	12	2 kb	14	11 kb	39.931885	74.5537	166	166	1120	1120
172.16.31.150	58220	44.228.249.3	80	3	186 bytes	86	2	120 bytes	1	66 bytes	133.654416	0.2926	3280	3280	1804	1804
172.16.31.150	64766	44.228.249.3	80	57	35 kb	62	25	7 kb	32	28 kb	59.417670	91.8337	589	589	2433	2433
172.16.31.150	54971	52.98.59.18	443	3	168 bytes	21	2	108 bytes	1	60 bytes	21.660093	0.0802	18	18	3979	3979
172.16.31.150	52348	52.98.59.18	443	346	303 kb	34	100	10 kb	246	354 kb	32.936590	22.1785	3428	3428	1574	1574
172.16.31.150	61586	52.98.59.18	443	47	23 kb	32	20	5 kb	27	18 kb	32.483320	22.6212	1894	1894	6217	6217
172.16.31.150	53217	52.182.143.214	443	308	264 kb	27	185	244 kb	123	20 kb	30.931139	24.4033	79	79	1413	1413
172.16.31.150	53128	52.182.143.214	443	21	10 kb	30	10	3 kb	11	7 kb	30.932472	24.3942	987	987	2378	2378
172.16.31.150	59568	52.182.143.214	443	30	12 kb	36	16	4 kb	14	8 kb	33.031773	22.2906	1428	1428	2810	2810
172.16.31.150	60467	52.182.143.214	443	21	10 kb	26	10	3 kb	11	7 kb	30.930842	1.2004	20	20	48	48

websites_track.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn=0&tcp.flags.ack=1

No.	Time	Source	Destination	Protocol	Length	Info
9250	60.408059	172.16.30.150	150.171.28.12	TLSv1.3	717	Change Cipher Spec, Client Hello (SNI=api.msn.com)
9254	60.415760	172.16.30.150	150.171.27.11	TCP	54	65344 → 443 [ACK] Seq=2701 Ack=37736 Win=65280 Len=0
9257	60.418068	172.16.30.150	150.171.27.11	TCP	54	65344 → 443 [ACK] Seq=2701 Ack=40656 Win=65280 Len=0
9260	60.420666	172.16.30.150	150.171.27.11	TCP	54	65344 → 443 [ACK] Seq=2701 Ack=43576 Win=65280 Len=0
9265	60.423321	172.16.30.150	150.171.27.11	TCP	54	65344 → 443 [ACK] Seq=2701 Ack=49416 Win=65280 Len=0
9269	60.424032	172.16.30.150	150.171.27.11	TCP	54	65344 → 443 [ACK] Seq=2701 Ack=53796 Win=65280 Len=0
9271	60.424139	172.16.30.150	150.171.27.11	TCP	54	65344 → 443 [ACK] Seq=2701 Ack=55256 Win=65280 Len=0
9274	60.424591	172.16.30.150	150.171.27.11	TCP	54	65344 → 443 [ACK] Seq=2701 Ack=58027 Win=65280 Len=0
9277	60.425080	172.16.30.150	150.171.27.11	TCP	54	65344 → 443 [ACK] Seq=2701 Ack=59695 Win=65280 Len=0
9279	60.428651	172.16.30.150	172.188.155.25	TCP	54	60696 → 443 [ACK] Seq=5146 Ack=5063 Win=65280 Len=0
9281	60.429907	172.16.30.150	172.188.155.25	TCP	54	60696 → 443 [ACK] Seq=5146 Ack=5906 Win=65024 Len=0
9285	60.435227	172.16.30.150	172.188.155.25	TCP	54	60696 → 443 [ACK] Seq=5146 Ack=6620 Win=64517 Len=0
9287	60.435436	172.16.30.150	20.42.65.84	TCP	54	58938 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=0
9289	60.436875	172.16.30.150	20.42.65.84	TCP	1494	58938 → 443 [ACK] Seq=1 Ack=1 Win=65280 Len=1440 [TCP PDU reassembled in 9290]
9290	60.436875	172.16.30.150	20.42.65.84	TLSv1.3	551	Client Hello (SNI=functional.events.data.microsoft.com)
9294	60.439122	172.16.30.150	150.171.28.12	TCP	54	64333 → 443 [ACK] Seq=2481 Ack=4480 Win=65280 Len=0
9297	60.439656	172.16.30.150	150.171.28.12	TCP	54	64333 → 443 [ACK] Seq=2481 Ack=6158 Win=65280 Len=0
9298	60.443818	172.16.30.150	150.171.28.12	TLSv1.3	128	Application Data
9303	60.476416	172.16.30.150	150.171.28.12	TCP	54	64333 → 443 [ACK] Seq=2555 Ack=6679 Win=64768 Len=0
9320	60.522899	172.16.30.150	44.228.249.3	TCP	54	52942 → 80 [ACK] Seq=875 Ack=4448 Win=65280 Len=0
9324	60.523356	172.16.30.150	44.228.249.3	TCP	54	64766 → 80 [ACK] Seq=410 Ack=1701 Win=65280 Len=0
9325	60.523386	172.16.30.150	44.228.249.3	TCP	54	52942 → 80 [ACK] Seq=875 Ack=7368 Win=65280 Len=0
9329	60.524179	172.16.30.150	44.228.249.3	TCP	54	64766 → 80 [ACK] Seq=410 Ack=4621 Win=65280 Len=0

Frame 9285: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF...
Ethernet II, Src: LiteonTechno_a3:5b:2b (d0:39:57:a3:5b:2b), Dst: Sonicwall_ee:46:3d (18:b1:69:ee:46:3d)
Destination: Sonicwall_ee:46:3d (18:b1:69:ee:46:3d)
Type: IPv4 (0x0800)
[Stream index: 8]
Internet Protocol Version 4, Src: 172.16.30.150, Dst: 172.188.155.25
Transmission Control Protocol, Src Port: 60696, Dst Port: 443, Seq: 5146, Ack: 6620, Len: 0

Specifies if this is an individual (unicast) or group (broadcast/multicast) address (eth.dst.ig), 1 bit

Packets: 12062 - Displayed: 5288 (43.8%) - Dropped: 0 (0.0%) Profile: Default

23°C Mostly sunny

websites_track.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

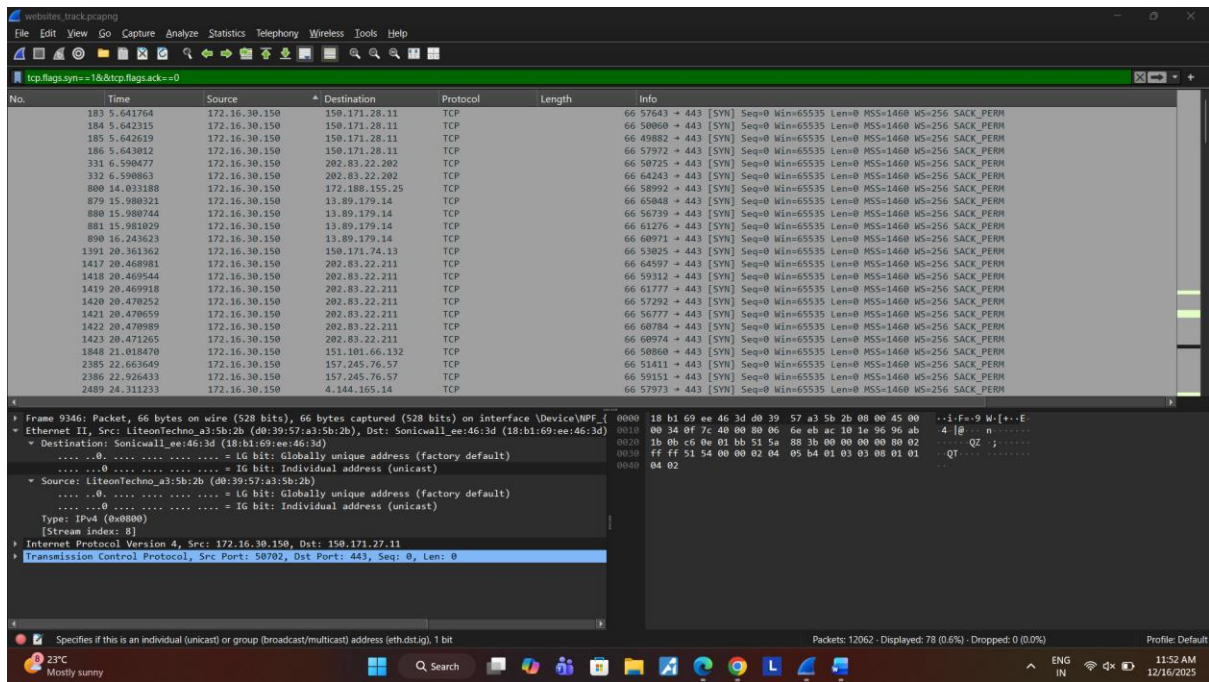
tcp.flags.syn=1&tcp.flags.ack=1

No.	Time	Source	Destination	Protocol	Length	Info
10211	79.468486	104.215.41.138	172.16.30.150	TCP	66	443 → 52415 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1440 SACK_PERM WS=1024
10010	75.299787	13.107.246.58	172.16.30.150	TCP	66	443 → 49533 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1440 SACK_PERM WS=512
10094	112.576201	13.107.6.158	172.16.30.150	TCP	66	443 → 51920 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
891	16.254472	13.89.179.14	172.16.30.150	TCP	66	443 → 56739 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
893	16.255558	13.89.179.14	172.16.30.150	TCP	66	443 → 65048 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
897	16.256021	13.89.179.14	172.16.30.150	TCP	66	443 → 61276 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
912	16.517986	13.89.179.14	172.16.30.150	TCP	66	443 → 60971 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
6378	39.763564	142.250.77.131	172.16.30.150	TCP	66	443 → 57346 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
10528	91.551045	142.251.220.99	172.16.30.150	TCP	66	443 → 52416 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
2840	30.651847	142.251.43.67	172.16.30.150	TCP	66	443 → 63105 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
9172	60.301761	150.171.27.11	172.16.30.150	TCP	66	443 → 65344 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
9354	61.009097	150.171.27.11	172.16.30.150	TCP	66	443 → 51105 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
9356	61.009729	150.171.27.11	172.16.30.150	TCP	66	443 → 50702 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
9556	64.146702	150.171.27.11	172.16.30.150	TCP	66	443 → 65498 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
10960	114.656099	150.171.27.11	172.16.30.150	TCP	66	443 → 52230 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
9145	60.215083	150.171.28.10	172.16.30.150	TCP	66	443 → 62912 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
187	5.754482	150.171.28.11	172.16.30.150	TCP	66	443 → 49882 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
188	5.754482	150.171.28.11	172.16.30.150	TCP	66	443 → 50060 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
195	5.766798	150.171.28.11	172.16.30.150	TCP	66	443 → 57043 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
196	5.766798	150.171.28.11	172.16.30.150	TCP	66	443 → 57972 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
8105	47.167666	150.171.28.11	172.16.30.150	TCP	66	443 → 59022 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
9202	60.362908	150.171.28.12	172.16.30.150	TCP	66	443 → 64333 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
1393	20.389155	150.171.74.13	172.16.30.150	TCP	66	443 → 53025 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM

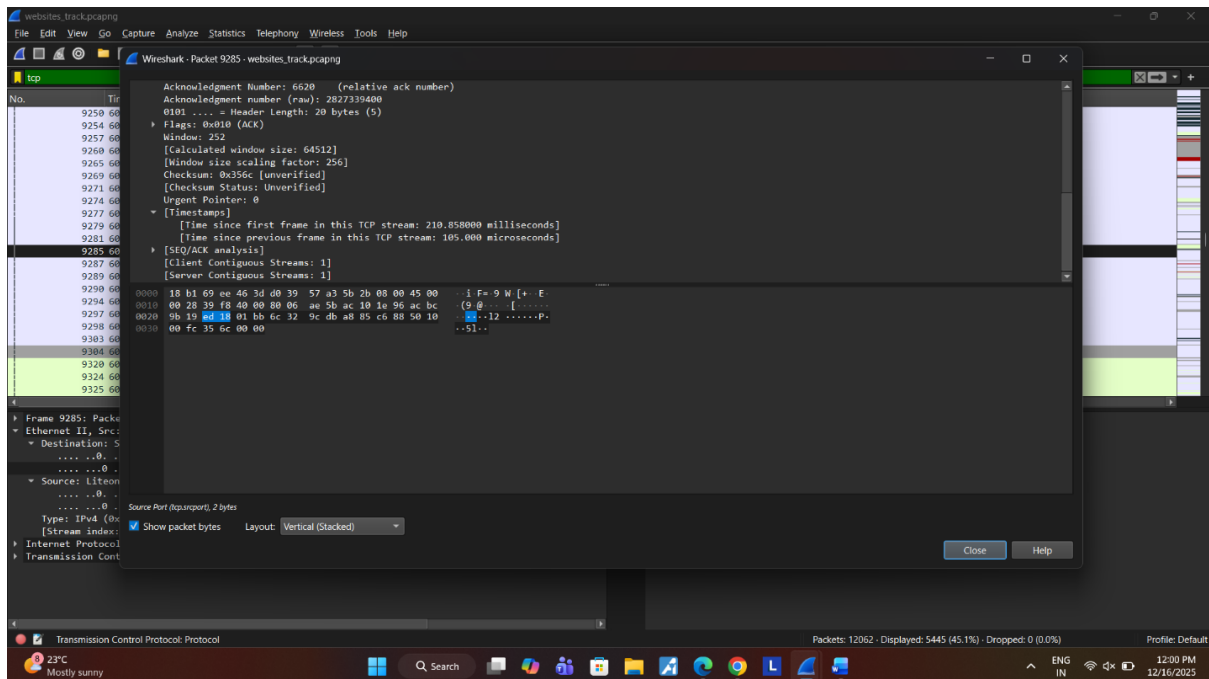
Frame 9286: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF...
Ethernet II, Src: Sonicwall_ee:46:3d (18:b1:69:ee:46:3d), Dst: LiteonTechno_a3:5b:2b (d0:39:57:a3:5b:2b)
Destination: LiteonTechno_a3:5b:2b (d0:39:57:a3:5b:2b)
Type: IPv4 (0x0800)
[Stream index: 8]
Internet Protocol Version 4, Src: 20.42.65.84, Dst: 172.16.30.150
Transmission Control Protocol, Src Port: 443, Dst Port: 58938, Seq: 0, Ack: 1, Len: 0

Packets: 12062 - Displayed: 78 (0.6%) - Dropped: 0 (0.0%) Profile: Default

23°C Mostly sunny



4. What is the time taken to establish TCP connection(s)? List this time taken value for each of the TCP connection(s).



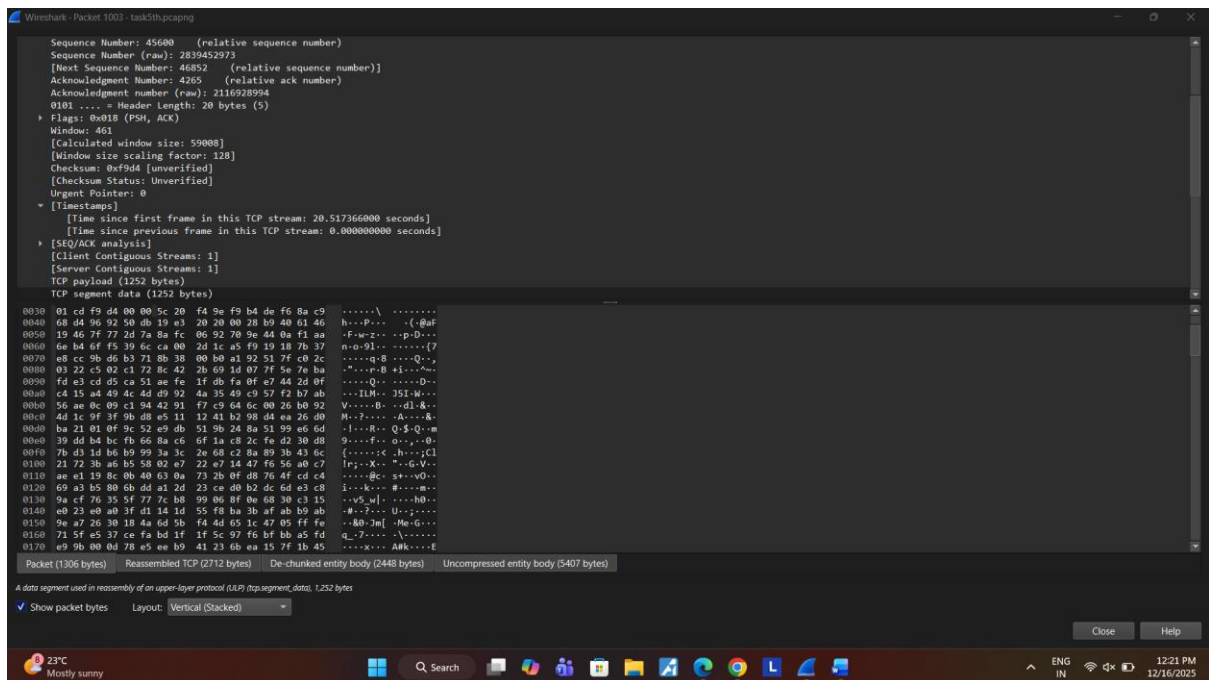
5. Browse the website by moving to various sub links, embedded objects listed in the site

The image shows a Wireshark packet capture of HTTP requests. The top pane displays a list of 167 requests, including GET requests for various resources like /categories.php, /artists.php, /cart.php, /listproducts.php, and /showimage.php. The bottom pane shows the details of the selected packet (No. 59), which is an Ethernet II frame from 172.16.31.26 to 239.255.255.250. The packet is an Internet Protocol Version 4, Src: 172.16.31.26, Dst: 239.255.255.250. The User Datagram Protocol (UDP) section shows Src Port: 51156, Dst Port: 1900. The Simple Service Discovery Protocol (SSDP) section shows the packet is a M-SEARCH request.

6. How many objects/files are downloaded:

The image shows a Wireshark packet capture of HTTP responses. The top pane displays a list of 1306 responses, including 200 OK (text/html) and 302 Found (text/html). The bottom pane shows the details of the selected packet (No. 71), which is an Ethernet II frame from 172.16.30.150 to 172.16.31.26. The packet is an Internet Protocol Version 4, Src: 172.16.30.150, Dst: 172.16.31.26. The Transmission Control Protocol (TCP) section shows Src Port: 80, Dst Port: 53950, Seq: 1461, Ack: 540, Len: 1287. The Hypertext Transfer Protocol (HTTP) section shows the response is 200 OK (text/html).

7. Make a detailed list including for each object/file downloaded what is the time taken for downloading the objects, the size of the object downloaded, object name, last modified time at the server.



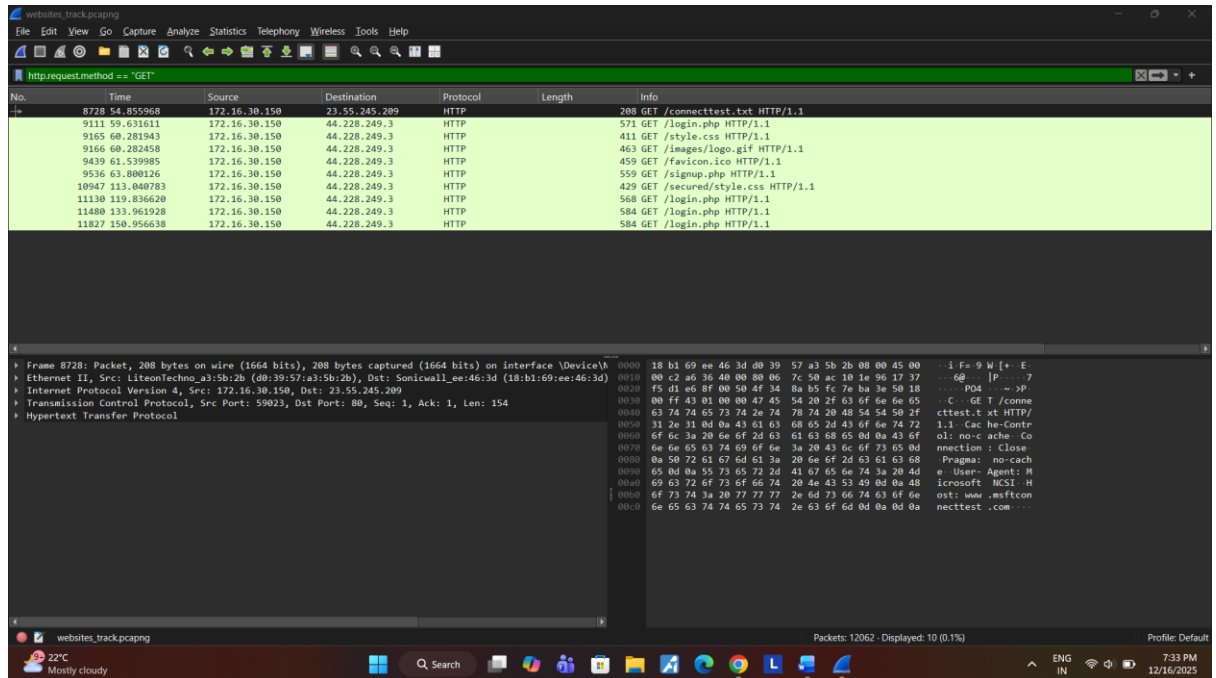
The image shows a Wireshark packet analysis window for a TCP segment. The packet list on the left shows a single packet (No. 1) of type TCP, length 1306 bytes. The packet details pane on the right shows the following information:

- Sequence Number: 45680 (relative sequence number)
- Sequence Number (raw): 2839452973
- [Next Sequence Number: 46852 (relative sequence number)]
- Acknowledgment Number: 4265 (relative ack number)
- Acknowledgment number (raw): 2116928994
- 0101 = Header length: 20 bytes (5)
- Flags: 0x018 (PSH, ACK)
- Window: 461
- [Calculated window size: 59008]
- Window size scaling factor: 128
- Checksum: 0xf9d4 (Unverified)
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- Timestamps
 - [Time since first frame in this TCP stream: 20.517366000 seconds]
 - [Time since previous frame in this TCP stream: 0.000000000 seconds]
- [SEQ/ACK analysis]
- [Client Contiguous Streams: 1]
- [Server Contiguous Streams: 1]
- TCP payload (1252 bytes)
- TCP segment data (1252 bytes)

The packet bytes pane at the bottom shows the raw data of the packet, starting with 0030 01 cd f9 d4 00 00 5c 20 f4 9e f9 b4 de f6 8a c9. The packet is shown in a vertical layout, and the packet bytes are displayed in hexadecimal and ASCII.

TASK 2

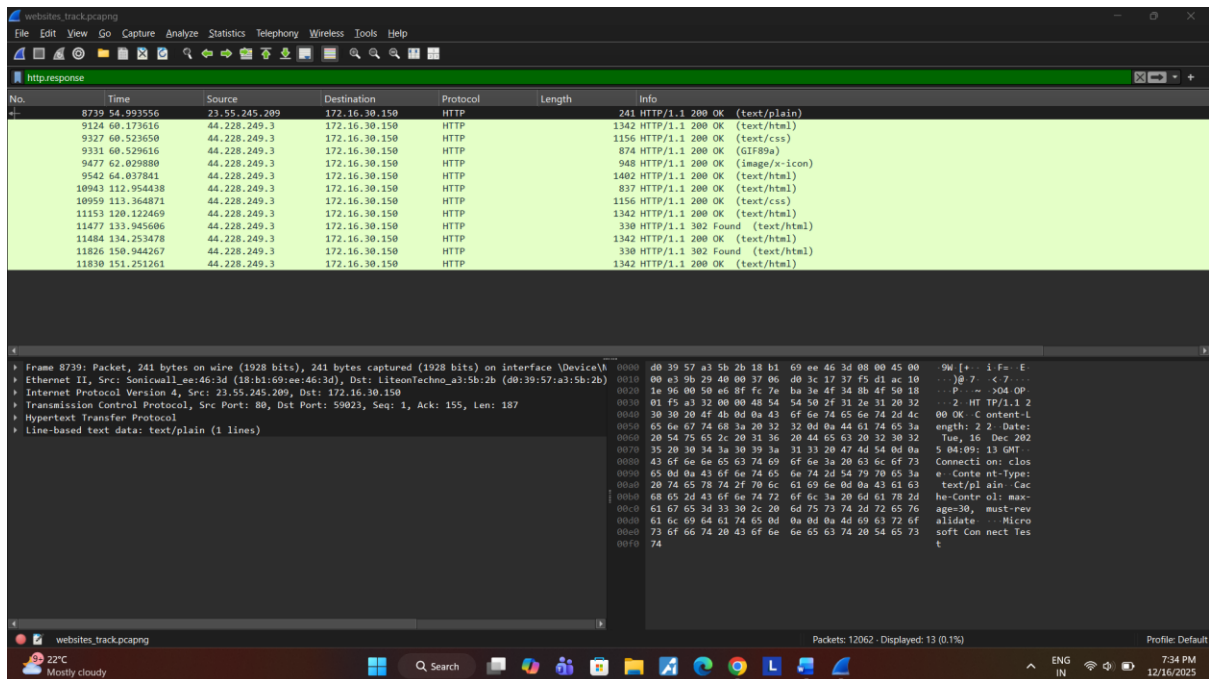
1. How many conditional GETs are sent by browser to the server :



The screenshot shows a Wireshark capture of network traffic. The top pane displays a list of packets, with the first 10 packets highlighted in green. These packets are all GET requests from 172.16.30.150 to 44.228.249.3. The bottom pane shows the details of the selected packet (No. 8728), which is a GET request for /connecttest.txt. The packet details show the Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Hypertext Transfer Protocol layers. The HTTP layer shows a 200 OK status and the content type is text/plain.

No.	Time	Source	Destination	Protocol	Length	Info
8728	54.855968	172.16.30.150	23.55.245.209	HTTP	208	GET /connecttest.txt HTTP/1.1
9111	59.631611	172.16.30.150	44.228.249.3	HTTP	571	GET /login.php HTTP/1.1
9165	60.281943	172.16.30.150	44.228.249.3	HTTP	411	GET /style.css HTTP/1.1
9166	60.282458	172.16.30.150	44.228.249.3	HTTP	463	GET /images/logo.gif HTTP/1.1
9439	61.539985	172.16.30.150	44.228.249.3	HTTP	459	GET /favicon.ico HTTP/1.1
9536	63.800126	172.16.30.150	44.228.249.3	HTTP	559	GET /signup.php HTTP/1.1
10947	113.840783	172.16.30.150	44.228.249.3	HTTP	429	GET /secured/style.css HTTP/1.1
11130	119.836620	172.16.30.150	44.228.249.3	HTTP	568	GET /login.php HTTP/1.1
11480	133.961928	172.16.30.150	44.228.249.3	HTTP	584	GET /login.php HTTP/1.1
11827	150.956638	172.16.30.150	44.228.249.3	HTTP	584	GET /login.php HTTP/1.1

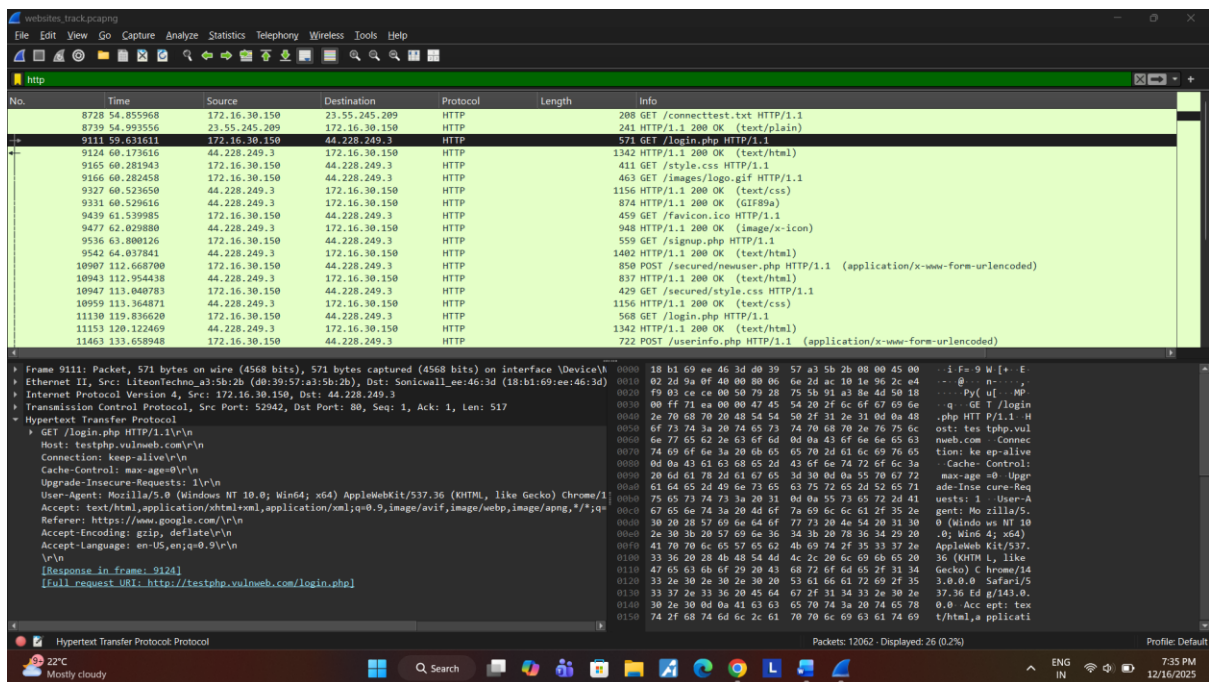
2. Make a list for each of the file/object downloaded, how many times the server sends the full contents of the respective file/object :



The screenshot shows a Wireshark capture of network traffic. The top pane displays a list of packets, with the first 13 packets highlighted in green. These packets are all HTTP responses from 23.55.245.209 to 172.16.30.150. The bottom pane shows the details of the selected packet (No. 8730), which is a 200 OK response for /connecttest.txt. The packet details show the Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Hypertext Transfer Protocol layers. The HTTP layer shows a 200 OK status and the content type is text/plain.

No.	Time	Source	Destination	Protocol	Length	Info
8730	54.993556	23.55.245.209	172.16.30.150	HTTP	241	HTTP/1.1 200 OK (text/plain)
9124	60.173616	44.228.249.3	172.16.30.150	HTTP	1342	HTTP/1.1 200 OK (text/html)
9327	60.523650	44.228.249.3	172.16.30.150	HTTP	1156	HTTP/1.1 200 OK (text/css)
9331	60.529616	44.228.249.3	172.16.30.150	HTTP	874	HTTP/1.1 200 OK (GIF89a)
9477	62.029880	44.228.249.3	172.16.30.150	HTTP	948	HTTP/1.1 200 OK (image/x-icon)
9542	64.037841	44.228.249.3	172.16.30.150	HTTP	1402	HTTP/1.1 200 OK (text/html)
10943	112.954438	44.228.249.3	172.16.30.150	HTTP	837	HTTP/1.1 200 OK (text/html)
10959	113.364871	44.228.249.3	172.16.30.150	HTTP	1156	HTTP/1.1 200 OK (text/css)
11153	120.122469	44.228.249.3	172.16.30.150	HTTP	1342	HTTP/1.1 200 OK (text/html)
11477	133.945606	44.228.249.3	172.16.30.150	HTTP	330	HTTP/1.1 302 Found (text/html)
11484	134.253478	44.228.249.3	172.16.30.150	HTTP	1342	HTTP/1.1 200 OK (text/html)
11826	150.944267	44.228.249.3	172.16.30.150	HTTP	330	HTTP/1.1 302 Found (text/html)
11830	151.251261	44.228.249.3	172.16.30.150	HTTP	1342	HTTP/1.1 200 OK (text/html)

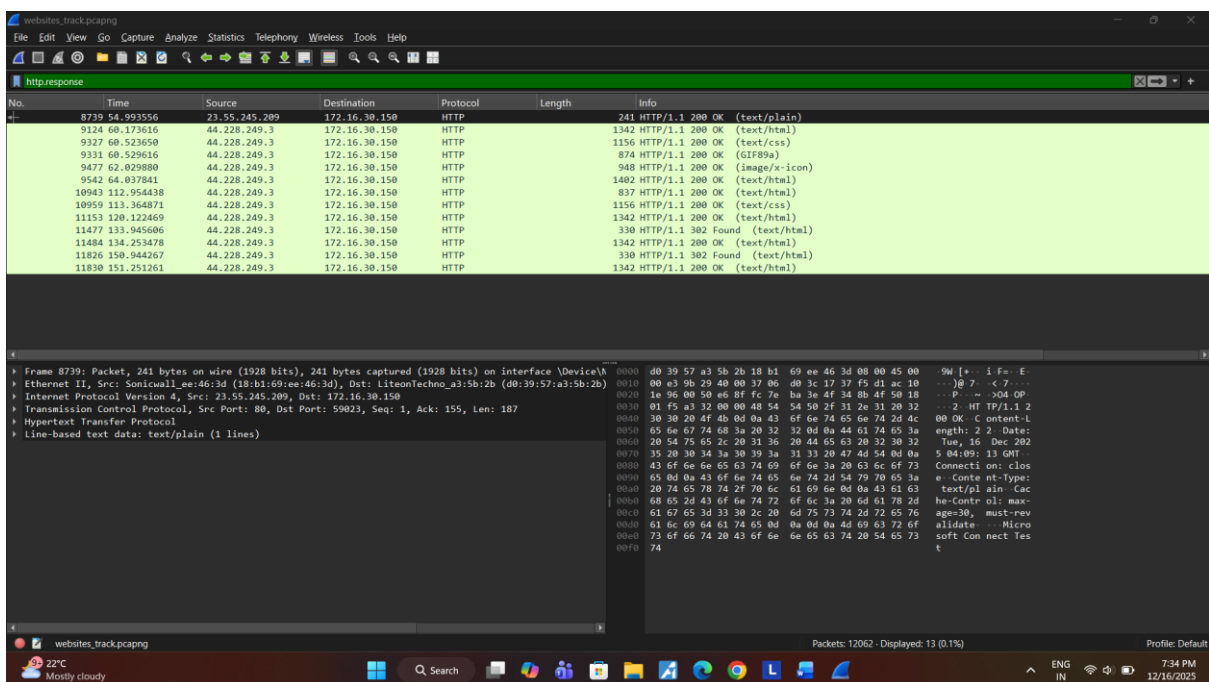
3. List the headers of HTTP which influence this functionality.

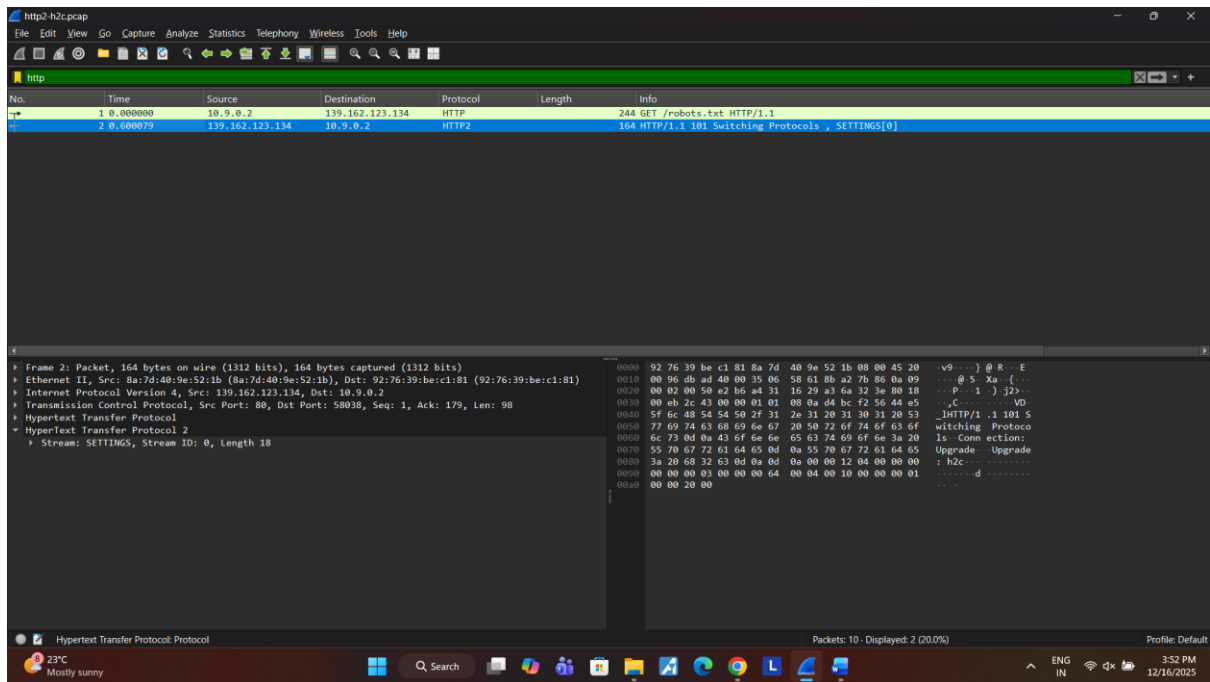


TASK 3

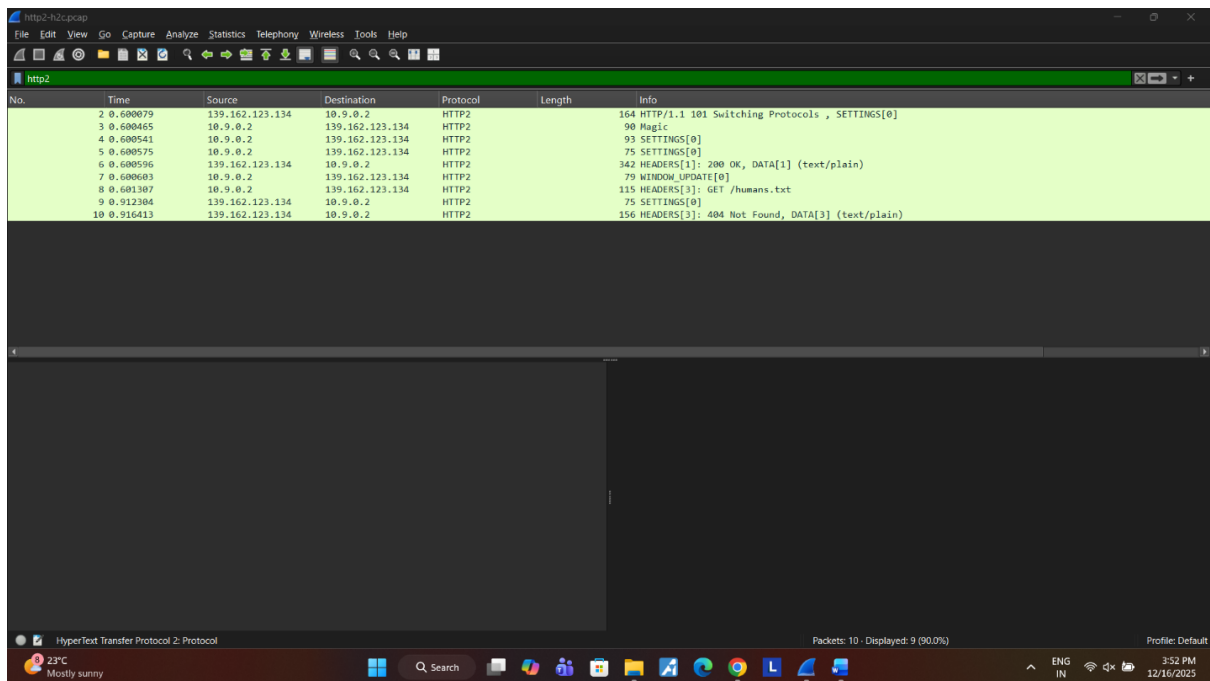
Analyze the attached HTTP/2 packet (http2-h2c.pcap: Included in zip file) capture using Wireshark to answer the following (Hint : Use Statistics->HTTP, HTTP2 windows).

1. How many HTTP/2 and HTTP/1.1 packets are present:





2.. How many HTTP/2 packets are exchanged between client and server here before the first object is fetched :



3. What main difference do you observe in headers of HTTP/2 packets displayed here, compared to the headers of HTTP/1.1 packets :

The image shows a Wireshark capture of HTTP/2 traffic. The packet list pane at the top shows several packets, with packet 10 selected. The packet details pane shows the structure of the selected packet, which is an HTTP/2 SETTINGS frame. The packet bytes pane shows the raw data of the packet.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
2	0.600079	139.162.123.134	10.9.0.2	HTTP2	164	HTTP/1.1 101 Switching Protocols , SETTINGS[0]
3	0.600465	10.9.0.2	139.162.123.134	HTTP2	90	magic
4	0.600541	10.9.0.2	139.162.123.134	HTTP2	93	SETTINGS[0]
5	0.600575	10.9.0.2	139.162.123.134	HTTP2	75	SETTINGS[0]
6	0.600596	139.162.123.134	10.9.0.2	HTTP2	342	HEADERS[1]: 200 OK, DATA[1] (text/plain)
7	0.600603	10.9.0.2	139.162.123.134	HTTP2	79	WINDOW_UPDATE[0]
8	0.601307	10.9.0.2	139.162.123.134	HTTP2	115	HEADERS[3]: GET /humans.txt
9	0.912304	139.162.123.134	10.9.0.2	HTTP2	75	SETTINGS[0]
10	0.916413	139.162.123.134	10.9.0.2	HTTP2	156	HEADERS[3]: 404 Not Found, DATA[3] (text/plain)

Packet Details (Frame 10):

- Frame 10: Packet, 156 bytes on wire (1312 bits), 156 bytes captured (1312 bits) on interface 0
- Ethernet II, Src: 8a:7d:48:9e:52:1b (8a:7d:48:9e:52:1b), Dst: 92:76:39:be:c1:81 (92:76:39:be:c1:81)
- Internet Protocol Version 4, Src: 139.162.123.134, Dst: 10.9.0.2
- Transmission Control Protocol, Src Port: 80, Dst Port: 58038, Seq: 1, Ack: 179, Len: 98
- Hypertext Transfer Protocol
- Hypertext Transfer Protocol 2
 - Stream: SETTINGS, Stream ID: 0, Length 18
 - Length: 18
 - Type: SETTINGS (4)
 - Flags: 0x00
 - 0... .. = Reserved: 0x0
 - = Stream Identifier: 0
 - Settings - Max concurrent streams : 100
 - Settings - Initial Window size : 1048576
 - Settings - Header table size : 8192

Packet Bytes:

0000 92 76 39 be c1 81 8a 7d 48 9e 52 1b 00 00 45 20 ... 0 R E
0010 00 96 db ad 40 00 35 06 58 61 8b a2 7b 86 0a 09 ... @ 5 Xa {
0020 00 02 00 50 e2 b6 a4 31 16 29 a3 6a 32 3e 80 18 ... P 1) j2s
0030 00 eb 2c 43 00 00 01 01 08 0a d4 bc f2 56 44 e5 ... C ... VD
0040 5f 6c 08 54 54 50 2f 31 2e 31 20 31 20 21 20 53 ... HTTP/1.1 101 5
0050 77 69 74 63 68 69 6e 67 20 50 72 6f 74 6f 63 6f ... witching Protoco
0060 6c 73 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 ... ls: Connection:
0070 55 70 67 72 61 64 65 0a 55 70 67 72 61 64 65 ... Upgrade: Upgrade
0080 3a 20 68 32 63 0d 0a 0a 00 00 12 04 00 00 00 ... : h2c
0090 00 00 00 03 00 00 00 64 00 04 00 10 00 00 01d ..
00a0 00 00 20 00